Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production engineering experience in C/C++, Python, and Linux systems, building fault-tolerant infrastructure, low-level automation, and distributed backend platforms on AWS. Strong background in reliability engineering, CI/CD, and root-cause diagnosis across Linux and cloud environments.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: C/C++, Java, Python, JavaScript, TypeScript, SQL, Go.

Systems & Infrastructure: Linux, embedded systems fundamentals, low-level device I/O, real-time debugging, fault-tolerant system design.

Backend: REST APIs, Node.js, Django, Flask, FastAPI, distributed systems.

Databases: PostgreSQL, MySQL.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Ansible, Bash, SonarQube.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** baseline across 53 runs on 20+ apps.
- Built device automation harness with stall detection and history-windowed decision loop; demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Diagnosed and resolved live infrastructure faults across Linux, EC2, and databases during **44 production environment upgrades** in 4 months, maintaining service reliability without ground-team escalation.
- Automated Teamcenter X infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Designed **Ansible-based Blue-Green deployment** framework with automated rollback, reducing production upgrade downtime by **50%** and enabling recovery without manual intervention.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, and GitLab CI/CD, improving fault tolerance and over-the-air style rollout safety.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** validation framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

SoundScape – Android Music Streaming App [Android SDK, Java, REST APIs, MediaPlayer] Dec 2021

- Developed end-to-end Android app with microservice-style separation for playback, notifications, and file operations; integrated Musixmatch API with local caching, cutting API calls by **70%**.

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset); designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).